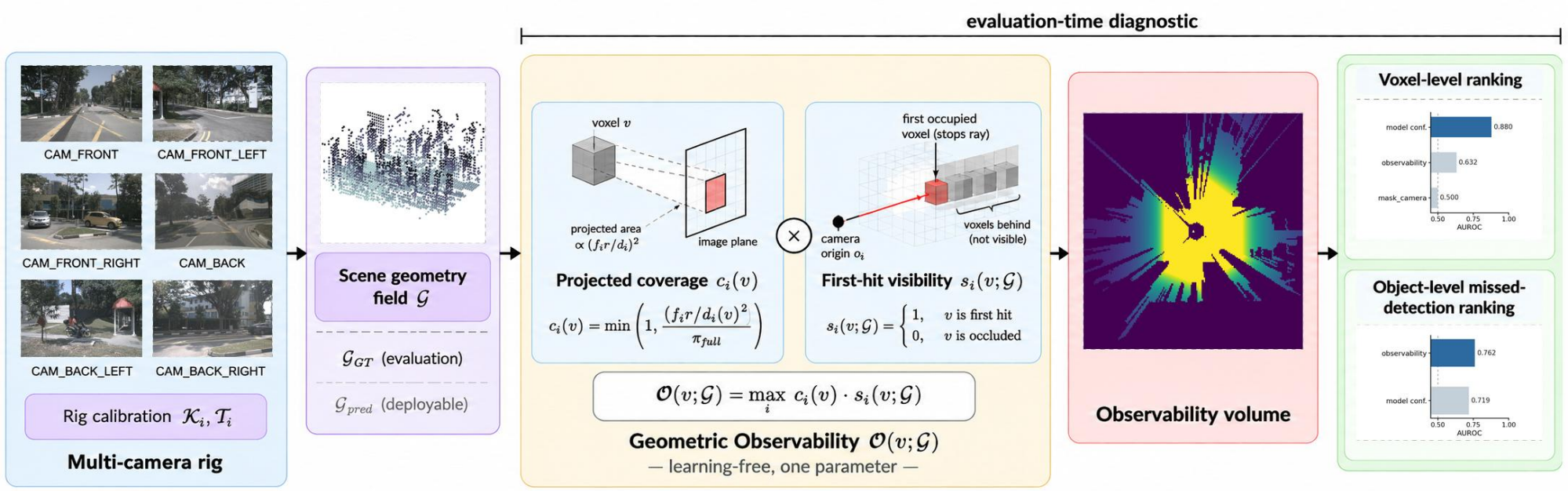

evaluation-time diagnostic
CAM_FRONT
CAM_FRONT_LEFT
CAM_FRONT_RIGHT
CAM_BACK
CAM_BACK_LEFT
CAM_BACK_RIGHT
Rig calibration $\mathcal{K}_i, \mathcal{T}_i$
Multi-camera rig
Scene geometry field $\mathcal{G}$
$\mathcal{G}_{GT}$ (evaluation)
$\mathcal{G}_{pred}$ (deployable)
voxel v
projected area $\propto (f_i r/d_i)^2$
image plane
Projected coverage $c_i(v)$
$c_i(v) = \min\left(1, \frac{(f_i r/d_i(v))^2}{\pi_{full}}\right)$
first occupied voxel (stops ray)
camera origin o_i
voxels behind (not visible)
First-hit visibility $s_i(v;\mathcal{G})$
$s_i(v;\mathcal{G}) = \begin{cases} 1, & v \text{ is first hit} \\ 0, & v \text{ is occluded} \end{cases}$
$\mathcal{O}(v;\mathcal{G}) = \max_i c_i(v) \cdot s_i(v;\mathcal{G})$
Geometric Observability $\mathcal{O}(v;\mathcal{G})$
— learning-free, one parameter —
Observability volume
Voxel-level ranking
model conf.
0.880
observability
0.632
mask_camera
0.500
0.50
0.75
1.00
AUROC
Object-level missed-detection ranking
observability
0.762
model conf.
0.719
0.50
0.75
1.00
AUROC